

Ibrahim Ali

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EDUCATION

- Vellore Institute of Technology** Chennai, India
B.Tech in Computer Science and Engineering; CGPA: 8.7 *July 2020 – May 2024*

EXPERIENCE

- Air India Limited** Kochi, India
Software Engineer *August 2024 – Present*
 - Multiple Excellence Awards (2024–2025)** — Recognized by delivery leaders, architects, and operations teams for distinct contributions, including the *Dream Team Award* for system reliability and cost optimization, the *High Flyer Award* for consistent high performance, and *AI IMPACCT Champion* recognitions for accountability and transformation during production issue resolution.
 - Migrated and scaled Air India's message delivery platform to handle **~1M+ notifications/day** with peak loads exceeding **50,000 requests/hour**, using **Azure Service Bus topics and Event Hub** for messaging, producer-consumer decoupling, replayable event processing, and fault-tolerant delivery.
 - Owned and scaled a real-time, event-driven delivery platform handling **~3,500+ flight events/day** and delivering **~500,000 notifications/day**, eliminating polling-based schedulers and improving reliability, timeliness, and compute efficiency.
 - Reduced query latency from **~7 seconds to under 50 ms** by eliminating nested lookups and introducing optimized in-memory mapping collections for high-frequency access paths.
 - Optimized notification processing architecture, reducing operational costs from **Rs.40L/month to under Rs.10L/month** through efficient event processing, channel optimization, and system consolidation.
 - Introduced a **WhatsApp-first notification strategy**, reducing SMS and robocall costs by **~60% per month** while improving delivery reliability and customer engagement.
 - Designed and developed an internal **MCP server in Python** with tool-based access to backend **Spring Boot** services, enabling safe, on-demand database queries and API calls to retrieve live system state, cutting investigation time from hours to minutes without direct production access.
 - Ensured production-grade code quality through PR reviews, SonarQube checks, unit testing, and GitHub security scanning, maintaining high reliability across frequent production releases.

PROJECTS

- Scheduler and PNR Notification System:**
 - Architected a serverless scheduling engine triggering time-based workflows for **100,000+** daily booking events and **400,000+** scheduled notification executions.
 - Engineered idempotent execution using state-based deduplication keys, eliminating duplicate notifications and reducing customer support tickets by **~20%**.
 - Integrated structured event parsing and downstream dispatch via REST APIs and message queues with reliable retry and recovery paths.
- Flight Disruption Management System (Conversational AI Platform):**
 - Initially built as a hackathon prototype using **Azure AI Foundry**, demonstrating automated passenger assistance during flight disruptions.
 - Designed an initial **4-agent architecture** (analysis, severity scoring, option generation, decisioning) for disruption impact evaluation and resolution generation.
 - Independently enhanced and production-hardened the system by introducing **deterministic state transitions** across **5 conversation stages**, improving reliability and traceability.
 - Expanded agent coordination logic and memory handling, reducing manual intervention during disruption spikes and improving response consistency.
 - Implemented the backend in **Python** with a **React.js** frontend, enabling extensibility for future airline disruption scenarios.

SKILLS

- Languages: Java, C/C++, Python, SQL, JavaScript
- Backend: Spring Boot, REST APIs, Microservices, Event-Driven Systems
- Distributed Systems: Messaging, Idempotency, Fault Tolerance, State Machines
- Cloud Platforms: Azure, AWS (S3, EC2, Lambda, CloudWatch)
- Containerization & CI/CD: Docker, Kubernetes, Azure Pipelines
- Databases: PostgreSQL, MySQL, MongoDB, Redis
- Foundations: Object-Oriented Design, Design Patterns, Data Structures